

Ayush Shukla

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SKILLS

Programming Languages: Python, C/cpp, JavaScript (ES6+), TypeScript

Frontend: React.js, HTML5, CSS3

Backend: Node.js, Express.js, Django, Flask, REST APIs

Databases: PostgreSQL, MongoDB, MySQL

Tools & Deployment: GitHub, Docker, VS Code, Render, Streamlit

EXPERIENCE

Full Stack Developer Intern

Jul 2025 – Aug 2025

Research Design & Standards Organisation (RDSO)

Lucknow, India

- Designed and delivered a production-ready full-stack soil suitability analysis tool actively used by railway and civil engineers, directly supporting infrastructure decisions across ongoing field projects.
- Engineered 10+ REST API endpoints using Django REST Framework; implemented soil classification algorithms, designed normalized PostgreSQL schemas, and achieved zero data integrity issues across all modules.
- Diagnosed and resolved critical bugs during integration testing; reduced API error rate by ~40% by collaborating with domain engineers to align output with field standards.

Summer Research Intern

June 2025

Maulana Azad National Institute of Technology (MANIT)

Bhopal, India

- Built 3 Python backend modules for an AI-powered ISRO space science educational platform; integrated ML inference pipelines that improved question-answering response accuracy by ~35%.
- Prototyped 3 interactive learning modules with dynamic AI-driven responses, increasing student engagement with space science content across the platform.

PROJECTS

Real-Time Chat Application | React, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.io

- Architected a production-grade full-stack chat app from scratch without Firebase/Supabase; built a custom WebSocket server using Socket.io, achieving <100ms message delivery latency for real-time messaging and live user presence tracking.
- Integrated Clerk auth with webhook handling and Express middleware; implemented cron jobs, raw file upload pipelines, and ImageKit for media transformations and CDN-backed image/video delivery, cutting media load time by ~50%.
- Delivered a responsive frontend with 5 themes, custom wallpapers, dark/light mode, and keyboard sound effects using Tailwind CSS and Hero UI, enhancing overall user experience.

Soil Suitability App and Website | Flutter, Django, PostgreSQL

- Built 8+ REST APIs for soil blanket thickness analysis for railway infrastructure evaluation; designed optimized PostgreSQL schemas and a Flutter frontend with real-time graphical visualization, reducing manual calculation time by ~60%.

Placement Portal Application | React, Node.js, PostgreSQL

- Developed a centralized full-stack placement portal supporting 1000+ students, enabling students to apply, recruiters to post jobs and browse profiles, and admins to manage the entire recruitment workflow.

AI Chatbot for Space Science Education | Python, Streamlit

- Built a multilingual AI chatbot with speech recognition and TTS supporting 3+ languages; integrated external APIs achieving ~90% question-answering accuracy for interactive ISRO/space science education.

Portfolio Website | HTML, CSS, JavaScript

- Developed a fully responsive portfolio with optimized performance, achieving sub-2s load time and cross-device compatibility across all screen sizes.

EDUCATION

Babasaheb Bhimrao Ambedkar University

Lucknow

Bachelor of Technology in Computer Science Engineering | CGPA: 8.5/10

2022 – 2026

EXTRA CURRICULAR

Student Placement Coordinator

2025 – 2026

Department of Computer Science Engineering, Babasaheb Bhimrao Ambedkar University

Lucknow

- Managing end-to-end placement drives for 100+ CSE students; acting as primary liaison between students and recruiting companies, coordinating interviews and guiding peers on resume building and interview preparation.

National Service Scheme (NSS)

2022 – 2023

Babasaheb Bhimrao Ambedkar University

Lucknow

- Contributed 120+ volunteer hours across community service drives, awareness campaigns, and rural outreach programs under NSS, Government of India.